

# ITERATION 2 - EVALUATION NOTES

## MotionCanvas

May 2026 | Evaluators: Breanna Gordon, Ammal Samatar, Katie Chen

### Purpose of This Iteration

To build a fully navigable prototype with a real interaction flow - onboarding, calibration, and a drawing canvas - and to validate that the core design concept (tremor reframed as expressive texture) was legible and functional.

### What Was Built

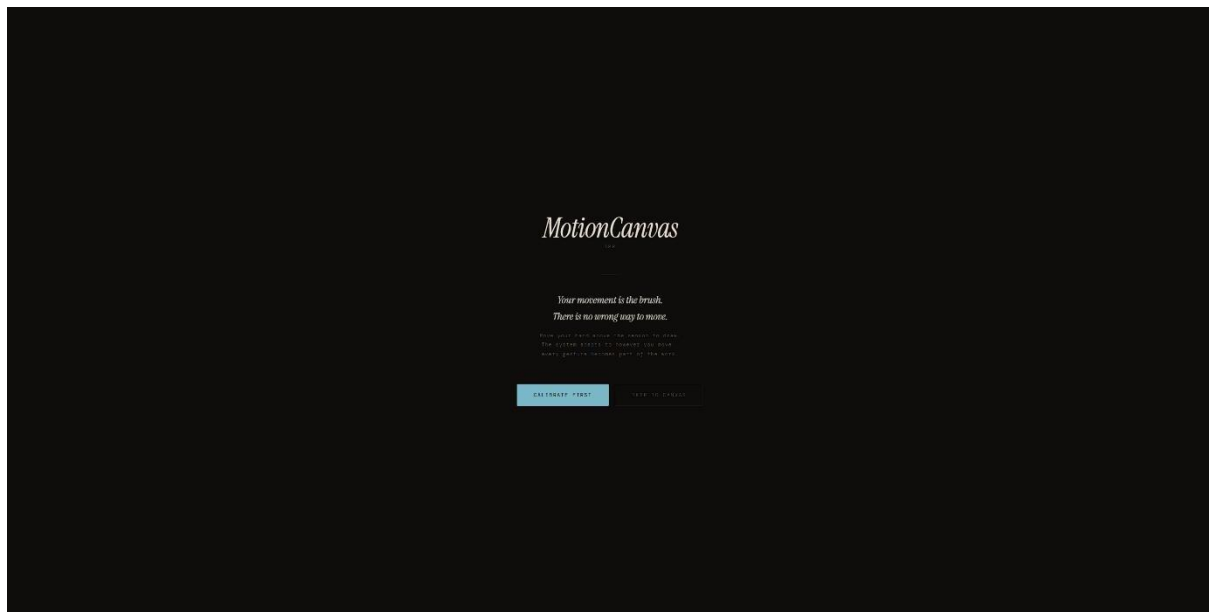
- Full p5.js rebuild with structured screen flow: Onboarding - Calibration - Canvas
- Four brush modes: Smooth, Particle, Scatter, Ink
- Colour grid (10 swatches) + custom colour picker
- Opacity and size sliders
- Functional tremor detection: dual-condition - high movement speed AND low positional displacement triggers tremor reframe
- Tremor reframe: cross-hatch marks drawn during detected tremor episodes
- Tremor intensity slider
- Simulate Tremor toggle for demonstration without requiring physical tremor
- Auto demo mode with speed control
- Light/dark mode toggle (partially implemented - canvas only)
- Mouse fallback



### Cognitive Walkthrough

A cognitive walkthrough traces a first-time user through the interface step by step, asking at each stage whether the user would know what to do and whether the system gives adequate feedback.

Step 1 - Onboarding Screen Task: Understand what the tool is and proceed to drawing.

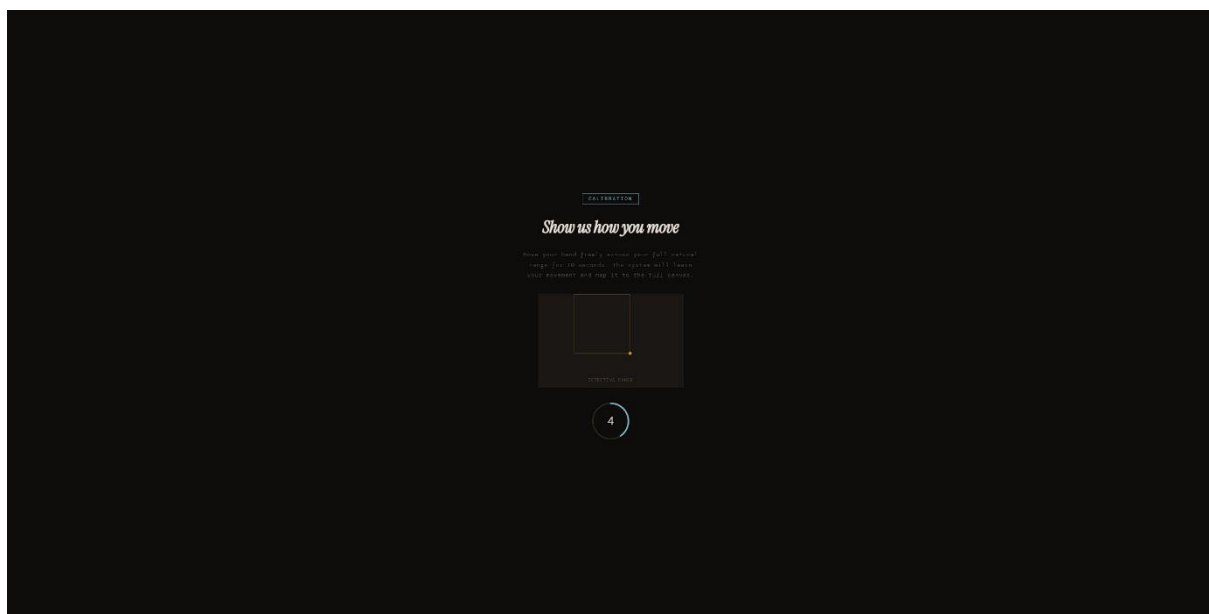


Pass: A new user can read the purpose and proceed without confusion.

Issue: There is no mention of the Leap Motion sensor on the onboarding screen. If the device is not plugged in, the user will only discover this after completing calibration.

Recommendation: Add a hardware check prompt or visible connection status on the onboarding screen.

Step 2 - Calibration Screen Task: Map personal movement range to the canvas.



Pass: The calibration screen includes a live preview box and an orange bounding rectangle that expands as the user moves. The 10-second countdown timer gives a time reference.

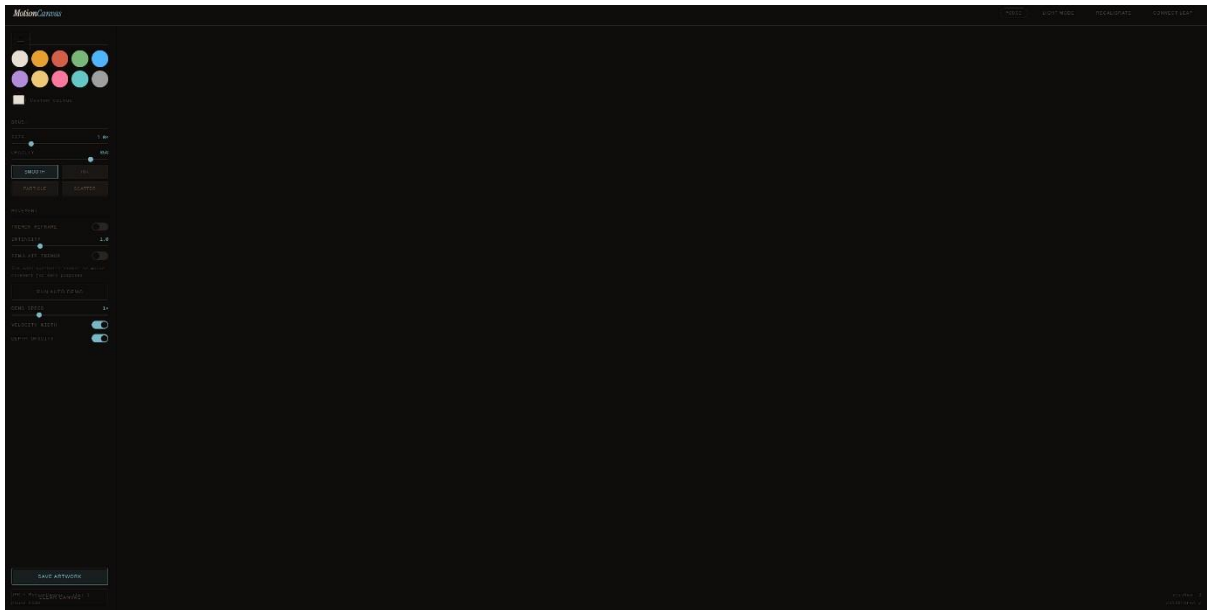
Issue 1: The "DETECTING RANGE" label does not explain what to do, only what is happening.

Issue 2: No confirmation state - when the countdown ends the screen transitions immediately with no visible success moment.

Issue 3: Users with a small or asymmetric movement range may be uncertain whether their range was sufficient.

Recommendation: Add a brief success state before transitioning to canvas.

Step 3 - Canvas and Drawing Task: Draw with the selected brush and discover the tremor reframe.



Pass: Drawing starts immediately. Brush mode, colour, and size controls are accessible in the side panel.

Issue 1: The side panel is information-dense. The Movement section is not clearly distinguished from brush and colour controls.

Issue 2: Tremor reframe fires and is visually distinct, but there is no explanatory label. A user unfamiliar with the concept may interpret it as a bug.

Issue 3: No undo. An unwanted mark cannot be reversed without clearing the whole canvas.

Recommendation: Add a label explaining tremor reframe. Add undo. Improve panel section grouping.

Step 4 - Tremor Reframe Discovery Task: Discover and understand what the tremor reframe does.

Pass: The simulate tremor button allows demonstration without physical tremor, essential for evaluation.

Issue: The concept is not explained anywhere in the UI. A new user sees cross-hatch marks appear with no context.

Recommendation: Add one line in the Movement panel explaining that tremor is being translated into texture rather than erased.

Strengths

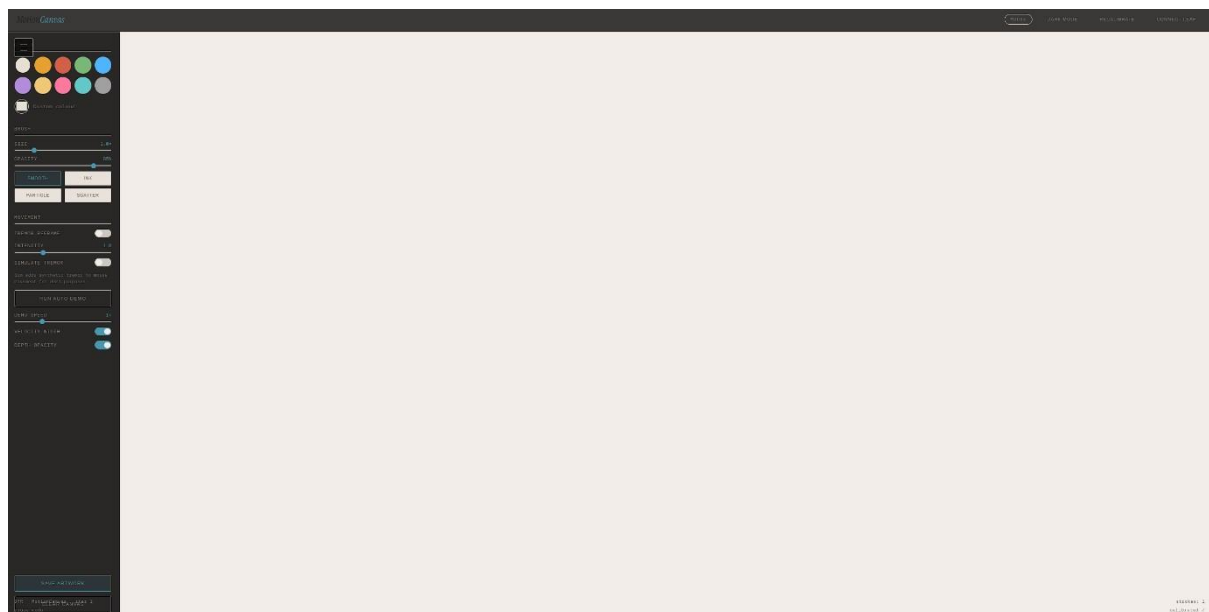
Tremor reframe is functional and visually clear - cross-hatch marks are distinct from all four brushes Simulated tremor and auto demo make the concept demonstrable without requiring physical tremor

p5.js gave significantly more expressive range - particle and scatter brushes support the movement-as-art concept

Warm visual identity is coherent and distinctive



Light/dark mode toggle was present as an accessibility accommodation, though full implementation was incomplete at this stage



### Key Limitations Leading Into Iteration 3

| Problem                                                         | Priority | Notes                                                |
|-----------------------------------------------------------------|----------|------------------------------------------------------|
| No undo                                                         | High     | Single most common frustration in any drawing tool   |
| Calibration lacks success state                                 | High     | Users cannot confirm their range was captured        |
| Tremor reframe not explained in-context                         | Medium   | Concept invisible unless user already knows about it |
| Leap status not dynamic                                         | Medium   | Tracking loss is silent to non-developer users       |
| Side panel grouping unclear                                     | Medium   | Movement controls blend visually with brush controls |
| Light mode applies only to canvas - panel and top bar stay dark | Medium   | Creates a visually inconsistent split                |
| Intensity slider has no numeric readout                         | Low      | Minor but useful for precision                       |

### Design Decisions for Iteration 3

Add undo - history stack to let users reverse unwanted marks

Calibration success state - confirmation before transitioning to canvas

Explicit tremor reframe label in the Movement panel

Dynamic Leap status indicator

Cleaner panel section grouping

Fix light mode - apply consistently across canvas, panel, and top bar

### Notes on Methodology

Evaluation was internal (group cognitive walkthrough). The auto demo and simulate tremor features were used for video documentation without requiring physical tremor.